

Gaussian Fourth Degree Function Report

Polynomial

$$f(q) = aq^4 + bq^3 + cq^2 + dq + e$$

Naive

$$g_{\text{naive}}(X) = Q^4a + Q^3b + Q^2c + Qd + e$$

$$\mathbb{E}[g_{\text{naive}}(X)] = aq^4 + 6aq^2\sigma^2 + 3a\sigma^4 + bq^3 + 3bq\sigma^2 + cq^2 + c\sigma^2 + dq + e$$

$$\begin{aligned}\text{Var}(g_{\text{naive}}(X)) = \sigma^2 & (16a^2q^6 + 168a^2q^4\sigma^2 + 384a^2q^2\sigma^4 + 96a^2\sigma^6 + 24abq^5 \\ & + 168abq^3\sigma^2 + 192abq\sigma^4 + 16acq^4 + 72acq^2\sigma^2 + 24ac\sigma^4 \\ & + 8adq^3 + 24adq\sigma^2 + 9b^2q^4 + 36b^2q^2\sigma^2 + 15b^2\sigma^4 + 12bcq^3 \\ & + 24bcq\sigma^2 + 6bdq^2 + 6bd\sigma^2 + 4c^2q^2 + 2c^2\sigma^2 + 4cdq + d^2)\end{aligned}$$

$$\begin{aligned}\text{MSE}(g_{\text{naive}}) = \sigma^2 & (16a^2q^6 + 204a^2q^4\sigma^2 + 420a^2q^2\sigma^4 + 105a^2\sigma^6 + 24abq^5 \\ & + 204abq^3\sigma^2 + 210abq\sigma^4 + 16acq^4 + 84acq^2\sigma^2 + 30ac\sigma^4 + 8adq^3 \\ & + 24adq\sigma^2 + 9b^2q^4 + 45b^2q^2\sigma^2 + 15b^2\sigma^4 + 12bcq^3 + 30bcq\sigma^2 \\ & + 6bdq^2 + 6bd\sigma^2 + 4c^2q^2 + 3c^2\sigma^2 + 4cdq + d^2)\end{aligned}$$

$$\text{Bias}(g_{\text{naive}}) = \sigma^2 (6aq^2 + 3a\sigma^2 + 3bq + c)$$

Unbiased

$$g_{\text{unb}}(X) = Q^4a + Q^3b - 6Q^2a\sigma^2 + Q^2c - 3Qb\sigma^2 + Qd + 3a\sigma^4 - c\sigma^2 + e$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^4 + bq^3 + cq^2 + dq + e$$

$$\begin{aligned}\text{Var}(g_{\text{unb}}(X)) = \sigma^2 & (16a^2q^6 + 72a^2q^4\sigma^2 + 96a^2q^2\sigma^4 + 24a^2\sigma^6 + 24abq^5 + 72abq^3\sigma^2 \\ & + 48abq\sigma^4 + 16acq^4 + 24acq^2\sigma^2 + 8adq^3 + 9b^2q^4 + 18b^2q^2\sigma^2 \\ & + 6b^2\sigma^4 + 12bcq^3 + 12bcq\sigma^2 + 6bdq^2 + 4c^2q^2 + 2c^2\sigma^2 + 4cdq + d^2)\end{aligned}$$

$$\begin{aligned}\text{MSE}(g_{\text{unb}}) = \sigma^2 & (16a^2q^6 + 72a^2q^4\sigma^2 + 96a^2q^2\sigma^4 + 24a^2\sigma^6 + 24abq^5 + 72abq^3\sigma^2 \\ & + 48abq\sigma^4 + 16acq^4 + 24acq^2\sigma^2 + 8adq^3 + 9b^2q^4 + 18b^2q^2\sigma^2 \\ & + 6b^2\sigma^4 + 12bcq^3 + 12bcq\sigma^2 + 6bdq^2 + 4c^2q^2 + 2c^2\sigma^2 + 4cdq + d^2)\end{aligned}$$

Comparison

$$\text{mean gap} = \sigma^2 (-6aq^2 - 3a\sigma^2 - 3bq - c)$$

$$\begin{aligned} \text{variance gap} = \sigma^4 & (-96a^2q^4 - 288a^2q^2\sigma^2 - 72a^2\sigma^4 - 96abq^3 - 144abq\sigma^2 - 48acq^2 \\ & - 24ac\sigma^2 - 24adq - 18b^2q^2 - 9b^2\sigma^2 - 12bcq - 6bd) \end{aligned}$$

$$\begin{aligned} \text{MSE gap} = \sigma^4 & (-132a^2q^4 - 324a^2q^2\sigma^2 - 81a^2\sigma^4 - 132abq^3 - 162abq\sigma^2 - 60acq^2 \\ & - 30ac\sigma^2 - 24adq - 27b^2q^2 - 9b^2\sigma^2 - 18bcq - 6bd - c^2) \end{aligned}$$

$$\text{Bias}(g_{\text{naive}})^2 = \sigma^4 (6aq^2 + 3a\sigma^2 + 3bq + c)^2$$